INCH-POUND

MIL-PRF-1/1389F
22 July 1999
SUPERSEDING
MIL-E-1/1389E(NAVY)
1 July 1975

# PERFORMANCE SPECIFICATION SHEET

## ELECTRON TUBE, CATHODE RAY
## TYPE 10ANP7

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: 10 inch, magnetic deflection, magnetic focus cathode ray tube with aluminized screen.

DIMENSIONS AND PIN CONNECTIONS: See figure 1.

ABSOLUTE RATINGS:

| Parameter: | Ef | Ec1 | Ec2 | Eb | Ehk | Rg1 | Alt |
|---|---|---|---|---|---|---|---|
| Unit: | V | V dc | V dc | V dc | V dc | Meg | ft |
| Maximum: | 6.9 | 0 | 450 | 11,000 | +125 | 1.5 | 20,000 |
| Minimum: | 5.7 | -200 | --- | 7,000 | -125 | --- | --- |
| Test condition: | 6.3 | Adj 1/ | 250 | 9,000 | --- | --- | --- |

See footnotes at end of table I.

GENERAL:

Qualification - Required. 5/


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Testing and inspection.

| Inspection | Method | Notes | Conditions | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| Qualification inspection | | | | | | | |
| Cathode illumination | 5216 | --- | | --- | --- | --- | --- |
| Direct-interelectrode capacitance | 1331 | --- | k to all<br>g1 to all | Ck<br>Cg1 | ---<br>--- | 5.0<br>6.3 | pF<br>pF |
| Electrode current (grid No. 2) | 5201 | --- | Ec1 = 0 | Ic2 | --- | 5.0 | µA dc |
| Pressure (implosion) | 1141 | --- | | --- | --- | --- | --- |
| Barometric pressure, reduced | 1002 | 7/ | | --- | --- | --- | --- |
| Vibration | 5111 | --- | | Width | --- | 2.0 | mm |
| Conformance inspection, part 1 | | | | | | | |
| Voltage breakdown | 5201 | --- | | --- | --- | --- | --- |
| Voltage breakdown (magnetic types) | 5201 | --- | | --- | --- | --- | --- |
| Gas ratio | 5206 | 6/ | | Gr | --- | 0.25 | --- |
| Neck straightness | 5101 | 2/ | | --- | --- | --- | --- |
| Screen and faceplate blemishes | 5106 | --- | | --- | --- | --- | --- |
| Modulation | 5223 | 1/ | Ib = 400 µA dc | ΔEc1 | --- | 40 | V dc |
| Spot position (magnetic deflection) | 5231 | --- | | Radius | --- | 12.5 | mm |
| Grid cutoff voltage | 5241 | --- | | Ec1 | 20 | 45 | V dc |
| Focus coil current | 5246 | --- | Ib = 100 µA dc;<br>D = 2.5 inch using JEDEC No. 127 focus coil, or equivalent | --- | 40 | 60 | mA dc |
| Aperture alignment | --- | 4/ | | --- | --- | 8 | mm |
| Grid No. 1 leakage current | 5251 | --- | | --- | --- | --- | --- |

See footnotes at end of table.



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TABLE I. Testing and inspection - Continued.

| Inspection | Method | Notes | Conditions | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | | | | Min | Max | |
| Conformance inspection, part 2 | | | | | | | |
| Heater current | 1301 | --- | | If | 270 | 330 | mA |
| Neck and bulb (magnetic types) | 5101 | --- | | --- | --- | --- | --- |
| Side terminal and base alignment | 5101 | --- | Halfway between 1 and 9 | --- | -10 | +10 | Degrees |
| Face tilt | 5101 | --- | | --- | --- | --- | --- |
| Stray light emission (conventional types) | 5216 | --- | Eb = 11,000 V dc; Ec2 = 450 V dc | --- | --- | --- | --- |
| Screens | 5221 | 3/ | | --- | --- | --- | --- |
| Line width A (magnetic deflection) | 5226 | --- | Ib = 100 µA dc | Width | --- | 0.33 | mm |
| Line width C (magnetic deflection) | 5226 | --- | Ib = 100 µA dc | Width | --- | 0.50 | mm |
| Heater-cathode leakage current | 5251 | --- | | --- | --- | --- | --- |
| Grid No. 2 leakage current | 5251 | --- | | --- | --- | --- | --- |
| Permanence of marking | 1105 | --- | | --- | --- | --- | --- |
| Conformance inspection, part 3 | | | | | | | |
| Life test | --- | --- | Group C; Eb = 11,000 V dc; Ec2 = 450 V dc; Ib = 60 µA dc | t | 500 | --- | hrs |
| Life-test end points: | --- | | | | | | |
| Line width A | 5226 | --- | Ib = 60 µA | Width | --- | 0.33 | mm |
| Modulation | 5223 | 1/ | Ib = 320 µA | ΔEc1 | --- | 40 | V dc |

See footnotes at top of next page.



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TABLE I. Testing and inspection - Continued.

1/ ΔEc1 shall be not greater than the value of grid cutoff voltage (see method 5241).

2/ Cylinder shall be 4 inches (101.60 mm) long and .910 inch (23.11 mm) maximum inside diameter.

3/ The screen characteristics shall be measured with constant beam energy of 0.24 watt defocused to a spot approximately 0.25 cm in diameter. The test conditions shall be anode voltage (relative to cathode) 9,000 volts and beam current 27 µA dc maximum. The screen characteristics shall comply with the following minimum limits: cB5 = 480 cB; G5: 1 = 4.

4/ The distance between the center of the unfocused, undeflected spot at low intensity (Ec1 near cutoff) and the center of the image of the masking aperture observed at high intensity of the unfocused, undeflected spot shall not exceed the limit specified herein. Ec1 shall not be held at zero for more than approximately 10 seconds to prevent damage to the screen.

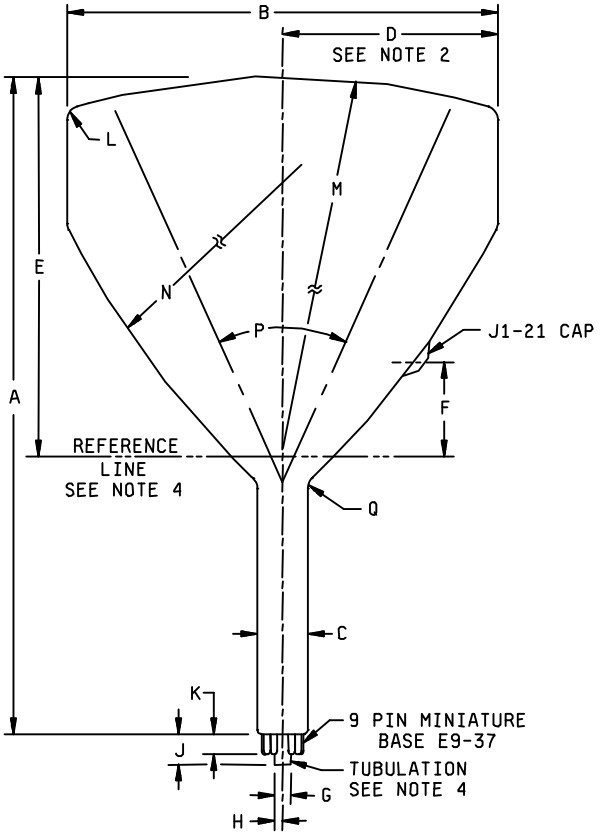
5/ The activity responsible for the qualified products list is the Defense Supply Center Columbus, Code VQE, 3990 E. Broad Street, Columbus, Ohio 43216-5000 and information pertaining to qualification of products may be obtained from that activity. Applications for qualification tests shall be made in accordance with "Provisions Governing Qualifications SD-6". (Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

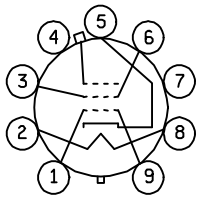
6/ This test is to be performed at the conclusion of the holding period.

7/ The low-pressure test shall be made with maximum voltages applied, and pressure equivalent to 20,000 feet (348 mmHg). Connections shall be made to the base pins in a manner that does not degrade the tube breakdown characteristics. Satisfactory operation shall be the absence of arc-over and harmful corona.



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BOTTOM VIEW

| PIN NO. | ELEMENT |
|---|---|
| 1 | GRID NO.1 |
| 2 | HEATER |
| 3 | GRID NO.2 |
| 5 | CATHODE |
| 6 | GRID NO.2 |
| 8 | HEATER |
| 9 | GRID NO.1 |
| CAP | ACCELERATOR |

| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | --- | 16.00 | --- | 406.40 |
| B | 10.437 | 10.563 | 265.10 | 268.30 |
| C | .810 | .896 | 20.57 | 22.76 |
| D | 4.500 | --- | 114.30 | --- |
| E | 9.233 | 9.483 | 234.52 | 240.87 |
| F | 1.915 | 2.415 | 48.64 | 61.34 |
| G | --- | .250 | --- | 6.35 |
| H | --- | .125 | --- | 3.18 |
| J | --- | .437 | --- | 11.10 |
| K | --- | .250 | --- | 6.35 |
| L | .437 Rad | | 11.10 Rad | |
| M | 42.000 Rad | | 1066.80 Rad | |
| N | 20.000 Rad | | 508.00 Rad | |
| P | 50° | | | |
| Q | .500 Rad | | 12.70 Rad | |

NOTES:
1. Reference line is determined by the point where JEDEC G-141 reference line gauge will stop.
2. Useful screen radius.
3. Dimensions without tolerance are for information only.
4. The tubulation diameter shall be less than the diameter of the socket hole and the tubulation shall not cant so as to bear on the sides of the socket hole.

FIGURE 1. Outline drawing of electron tube type 10ANP7.



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Custodians:
Navy - EC
DLA - CC

Review activities:
Navy - AS, CG, MC, OS, SH

Preparing activity:
DLA - CC

(Project 5960-3547-03)



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